

CHAPTER THREE

CIRCUIT MODELING

3.1 CASCADED H-BRIDGE INVERTER

The cascaded H-bridge inverter, as discussed earlier, makes use of two H-bridge, for five level modulation schemes. This increase in modulation level is to reduce the harmonic distortion and improve efficiency of the inverter system. To determine the number of H-bridge to be used for an n^{th} level inverter, the following equation is used:

$$\text{number of H - bridge} = \frac{m-1}{2} \quad \dots (3.1)$$

Where m is the level of the modulation scheme, which in this case is five levels. Thus:

$$\text{number of H - bridge} = \frac{5-1}{2} = 2$$

Thus, the number of H-bridges is two (2) as stated earlier. It should be noted that each H-bridge contains four (4) switches, making a total of eight (8) switches for the system. The eight switches will need eight different switching signals. Care should be taken to avoid turning on two switches on the same leg ON at a particular time to avoid shorting the supply voltage. The H-bridge configuration and switching scheme for the eight switches in the H-bridge system is discussed below.

3.1.1 THE H-BRIDGE

As stated above, two H-bridge systems connected in cascade will be used in this simulation. The system comprises eight switches which have different switching signals as will be discussed later. The Simulink diagram of the H-bridge system is as shown below. The dual H-bridge used for the multilevel inverter have separate DC sources usually smaller than the DC source used for a single

H-bridge. Cascaded H-Bridge configuration has recently become very popular in high-power AC supplies and adjustable-speed drive applications. A cascade multilevel inverter consists of a series of H-bridge (single-phase full bridge) inverter units. Each H-bridge unit has its own dc source. Each SDC (separate D.C. source) is associated with a single-phase full-bridge inverter. The ac terminal voltages of different level inverters are connected in series. Fig. 3.1 shows a single-phase structure of a cascaded H-bridge inverter with separate D.C. sources.

Through different combinations of the four switches, $S1-S4$, each converter level can generate three different voltage outputs, $+ V_{dc}$, $- V_{dc}$ and zero. To obtain

$+ V_{dc}$, switches $S1$ and $S4$ are turned ON. On turning ON $S2$ and $S3$ together, we get the output $-V_{dc}$. On turning the switches $S1$ and $S2$ together or $S3$ and $S4$ together or $S1, S2, S3, S4$ simultaneously, we get the output 0. The AC outputs of different full-bridge converters are connected in series such that the synthesized voltage waveform is the sum of the individual converter outputs. In this topology, the number of output-phase voltage levels is defined by $M=2N+1$, where 'M' is the no of levels and 'N' is the number of DC sources. The switching signals are sent to the Mosfet gates, in which the DC source is connected to its drain. This switching signal (also gating pulses), with the Mosfet turns the DC voltage into an AC voltage. Thus, for a two-source inverter, V_o can be any of the five levels $+ 2V_{dc}$, $+ V_{dc}$, 0, $- V_{dc}$, or $- 2V_{dc}$.

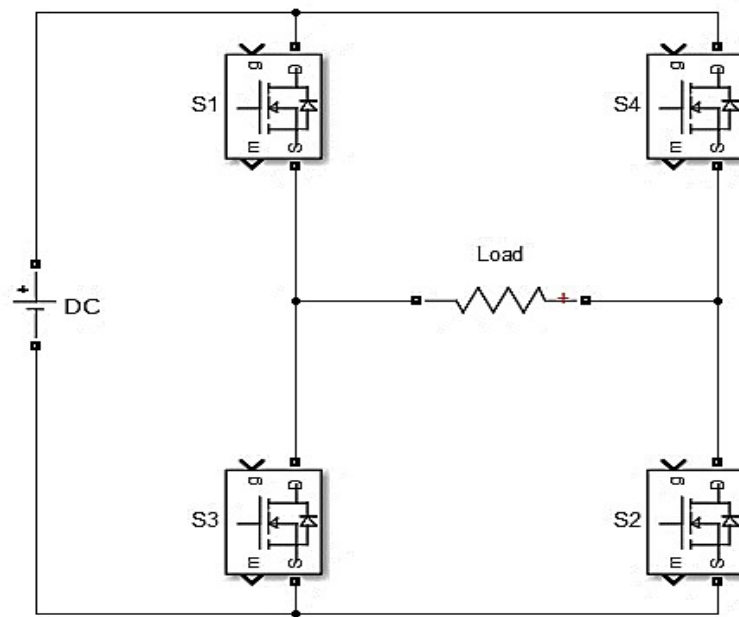


Fig. 3.1: Single H-bridge System with resistive load

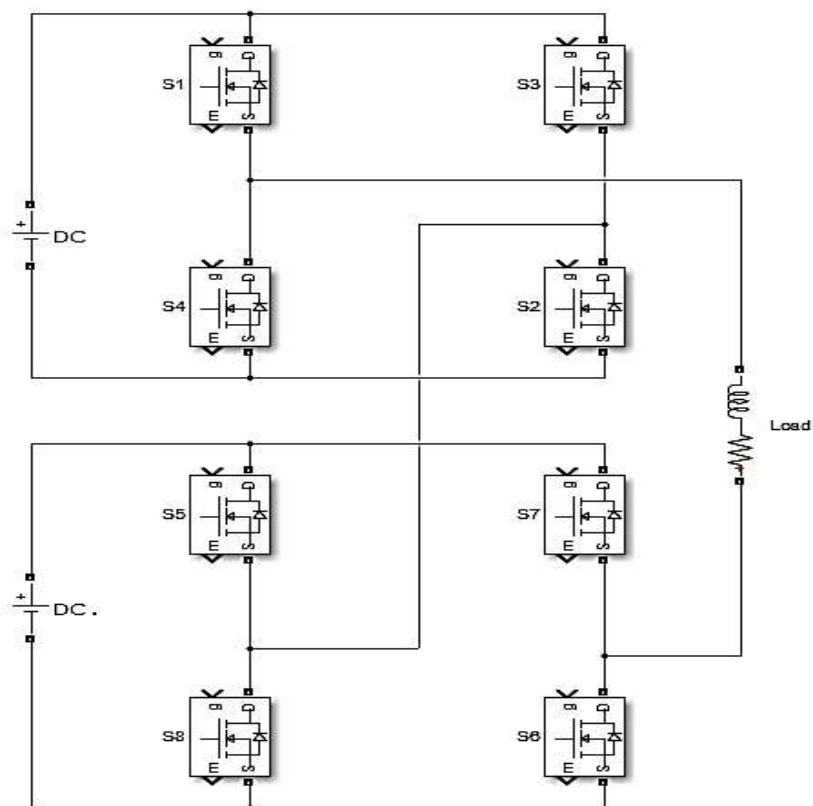


Fig. 3.2: Cascaded H-bridge Inverter with RL load

3.1.2 OUTPUT VOLTAGE LEVELS

The output voltage of a cascaded H-bridge five level inverter is not the same as that of an inverter with a single H-bridge. As stated earlier, the voltage level obtained is usually a sum of the two separate DC sources used for the two H-bridges. The expression for finding output voltage level is given as:

$$V_{ol} = 2N_{VDC} + 1 \quad \dots (3.2)$$

Where N_{VDC} and V_{ol} are the number of DC sources to be used and output voltage level respectively.

In this work, the cascaded H-bridge makes use of two separate DC sources, thus $N_{VDC} = 2$. From the expression above, the output voltage level is:

$$V_{ol} = (2 * 2) + 1 = 5$$

The table below shows the output voltage levels of the cascaded H-bridge. Let V_{o1} , V_{o2} and V_{ot} denote the output of the first H-bridge, second H-bridge and total output respectively.

Table 3.1: Output voltage levels of a single phase cascaded H-bridge five level inverter

No	V_{o1}	V_{o2}	V_{ot}
V_1	$+ V_{DC}$	$+ V_{DC}$	$+ 2V_{DC}$
V_2	$+ V_{DC}$	0	$+ V_{DC}$
V_3	0	0	0
V_4	0	$- V_{DC}$	$- V_{DC}$
V_5	$- V_{DC}$	$- V_{DC}$	$- 2V_{DC}$

From table 3.1 above, it can be observed that with a five-level cascaded H-bridge inverter with two equal and separate DC sources, the total output obtained at the load point is the sum of the two DC sources.

3.1.3 SWITCHING SCHEME

The sinusoidal PWM technique is very popular for industrial converters. Fig. 3.3 shows the general principle of SPWM, where an isosceles triangle carrier wave of frequency f_c , is compared with the fundamental frequency f sinusoidal modulating wave, and the points of intersection determine the switching points of power devices. Carriers used in multilevel inverters may be vertically shifted or horizontally shifted. Advantage of horizontally shifted carriers scheme is that each module is switched ON and OFF with a constant number of times per period, independent of magnitude of the generated voltage. But a vertically shifted carrier scheme can be more easily implemented on any digital controller.

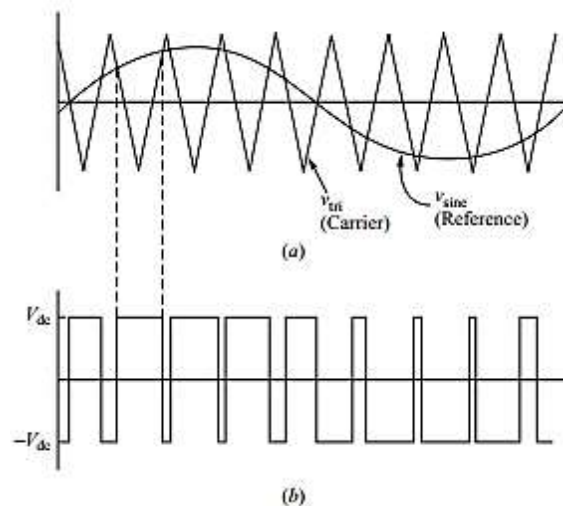


Fig. 3.3: SPWM principle (a) Sinusoidal reference and triangular carrier (b)
Output waveform

The switching is done in such a way that no two complementary switches are ON at the same time (such as $S1$ and $S4$) to avoid shorting the supply. To

achieve this, the switching signals are modulated using different switching (or delay) angle α , resulting in a bridge and total output as shown in fig. 3.4 below.

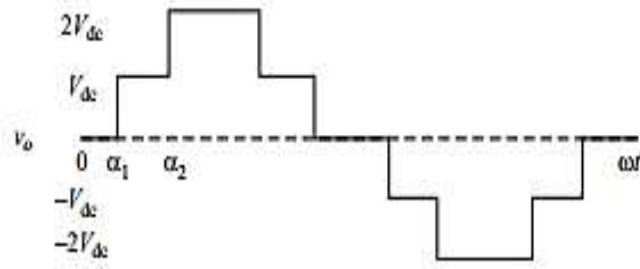


Fig. 3.4: Switching delay angle inverter output

The Fourier series for the total output voltage V_o for the five level two source inverters under study has only the odd-numbered harmonics and is given as:

$$V_o(t) = \frac{4V_{dc}}{\pi} \sum_{n=1,3,5,7...}^{\infty} \left[\cos(n\alpha_1) + \cos(n\alpha_2) \right] \frac{\sin(n\omega_o t)}{n} \quad \dots (3.3)$$

The Fourier coefficients for this series is given as:

$$V_n = \frac{4V_{dc}}{n\pi} \sum_{n=1,3,5,7...}^{\infty} \left[\cos(n\alpha_1) + \cos(n\alpha_2) \right] \quad \dots (3.4)$$

The modulation index M_i , is the ratio of amplitude of the fundamental frequency of the sinusoidal signal and the carrier triangular waveform, and is given as:

$$M_i = \frac{A_{sine}}{A_{tri}} \quad \dots (3.5)$$

It can also be said to be the ratio of the amplitude of the fundamental frequency component of V_o to the amplitude of the fundamental frequency component of a square wave of amplitude $2V_{dc}$, which is $2\left(\frac{4V_{dc}}{\pi}\right)$.

$$M_i = \frac{V_1}{2\left(\frac{4V_{dc}}{\pi}\right)} = \frac{\cos(\alpha_1) + \cos(\alpha_2)}{2} \quad \dots (3.6)$$

The delay angles, α_1 and α_2 , of the switching signals are calculated as follows:

$$\alpha_i = i * \frac{180^\circ}{m} \quad \dots (3.7)$$

Where:

$$i = 1, 2 \dots (m - 1)/2$$

m is level of switching output which in our case is five (5) level

α is the switching (delay) angle

Since this work is based on a five-level inverter, the α_i has a maximum of eight values of which only α_1 and α_2 according to the expression $(m - 1)/2$. The rest will be deduced by adding or subtracting from 180° or 360° .

$$\alpha_1 = 1 * \frac{180^\circ}{5} = 36^\circ$$

$$\alpha_2 = 2 * \frac{180^\circ}{5} = 72^\circ$$

The first and second switching angles are 36° and 72° respectively. It should be noted that the subscript attached to the delay angles indicate the switches at which they are applied. The table below shows the eight switches and their switching angles.

Table 3.1: Switching angles of the cascaded H-bridge

Switch	Angle α
S_1	36°
S_2	72°
S_3	$180^\circ - 72^\circ = 108^\circ$
S_4	$180^\circ - 36^\circ = 144^\circ$
S_5	$180^\circ + 36^\circ = 216^\circ$
S_6	$180^\circ + 72^\circ = 252^\circ$
S_7	$360^\circ - 72^\circ = 288^\circ$
S_8	$360^\circ - 36^\circ = 324^\circ$

These conduction angles are in accordance with figure 3.4 above, which shows the output waveform of the inverter. In this simulation, the sine wave frequency is set at 50Hz and the triangular carrier frequency set at 2KHz . This high frequency is due to the fact that the finer the pulses, the better the approximation of the output is closer to a pure sine wave.

The switching sequence of the H-bridge must be in such a way that no two switches in the same leg (say, S_1 and S_4) should be ON at the same time to avoid shorting the power supply. Table 3.2 shows the switching sequence of the five level inverters eight switches ($S_1 - S_8$).

Table 3.2: Switching sequence of the eight switches

V_{ot}	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8
V_1	1	0	0	1	1	0	0	1
V_2	1	0	0	1	1	0	1	0
V_3	1	0	1	0	1	0	1	0

V_4	1	0	1	0	0	1	1	0
V_5	0	1	1	0	0	1	1	0

Taking a good look at table 3.1, it can be seen that switches in the same leg have complementary switching signals (say, $S_1 = \overline{S_2}$, $S_4 = \overline{S_3}$, and so on). This is to avoid the situation of turning the two switches ON at the same time, causing a short in the DC supply.

3.2 MATLAB SIMULATION

The five-level cascaded H-bridge inverter was simulated using Simulink environment in MATLAB (R2014a). The cascaded B-bridge was implemented as shown in figure 3.5.

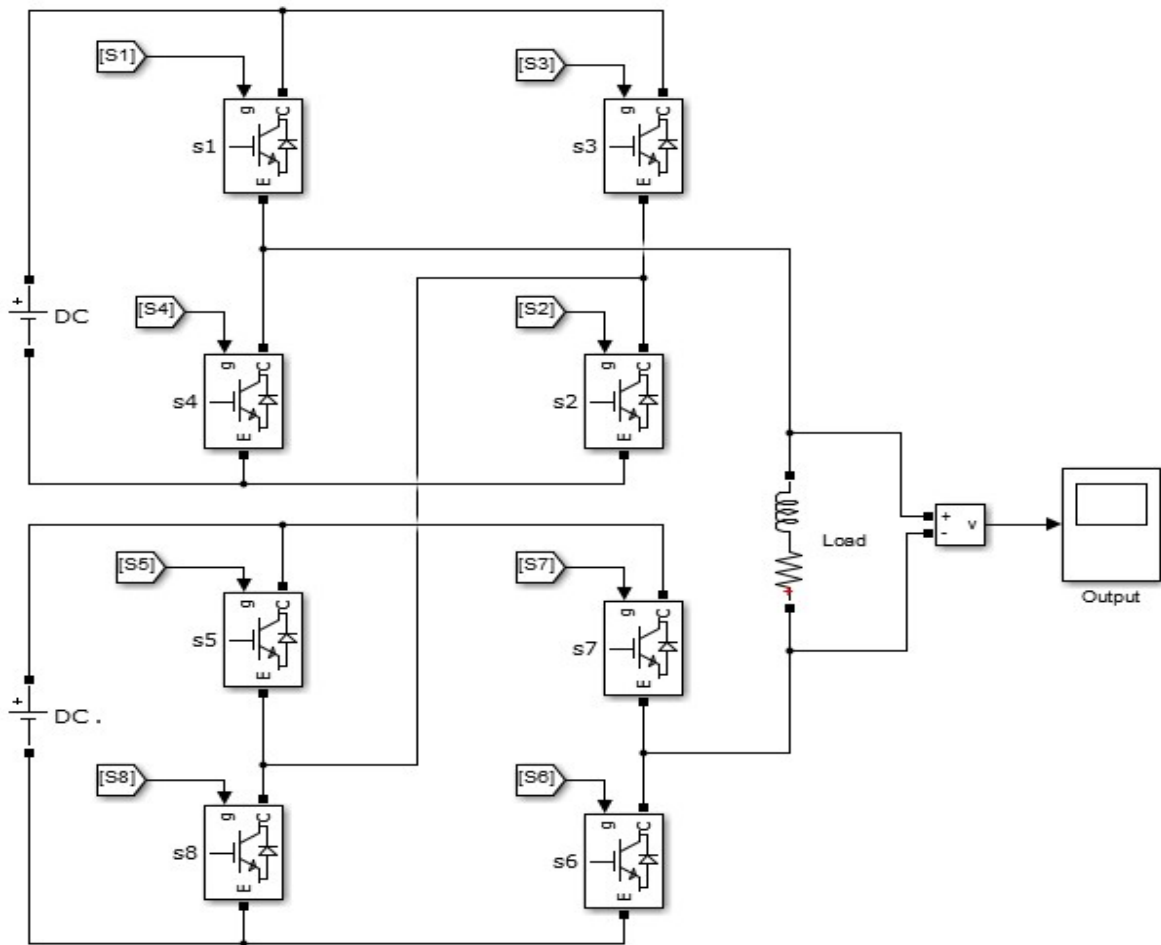


Fig. 3.5: Cascaded H-bridge System for the five-level inverter

The switching pulses are implemented as shown below:

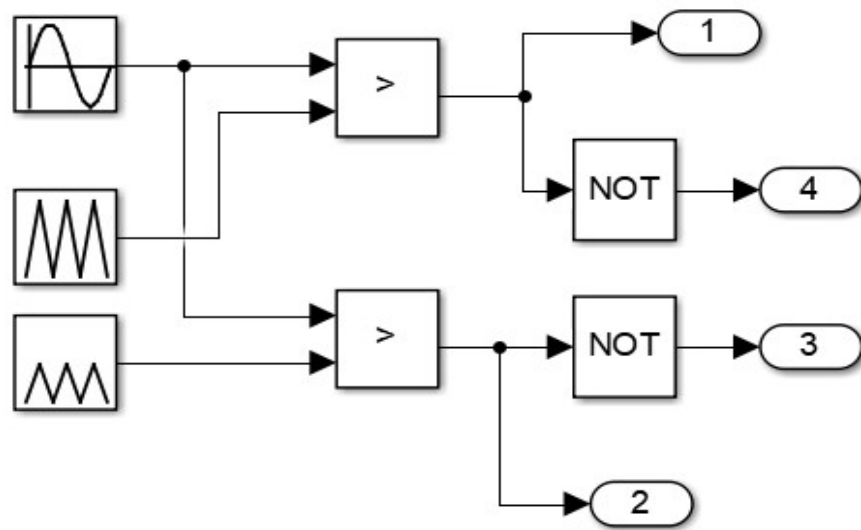


Fig. 3.6: SPWM switching pulses for switches $S1 - S4$

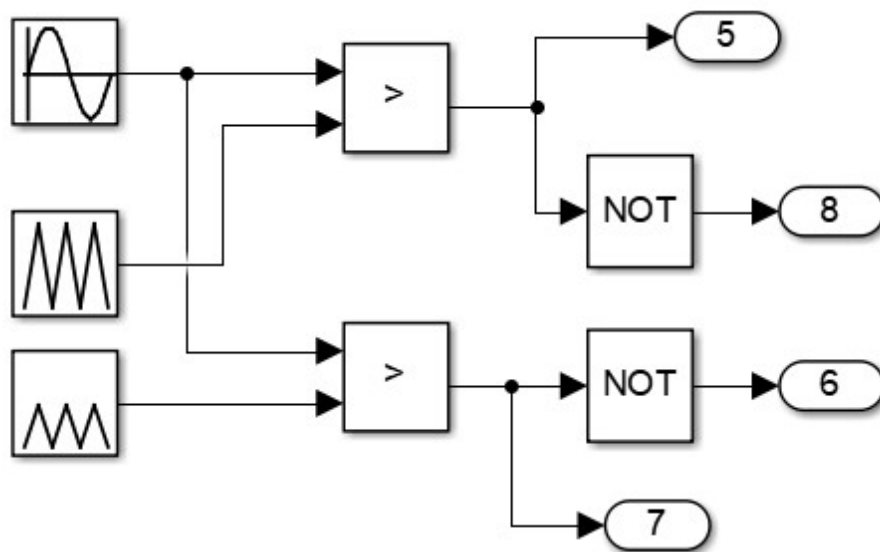


Fig. 3.7: Switching Pulse for switches $S5 - S8$

The overall inverter system is as shown in figure 3.8 below.

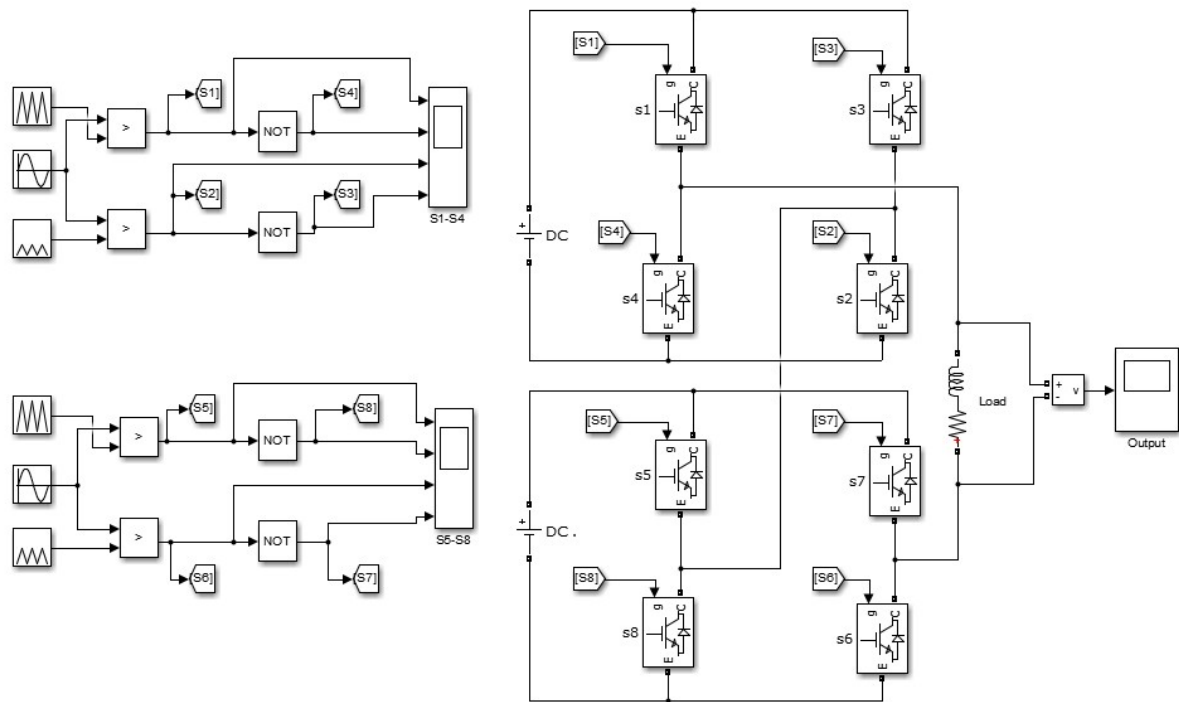


Fig. 3.8: Overall model of the system

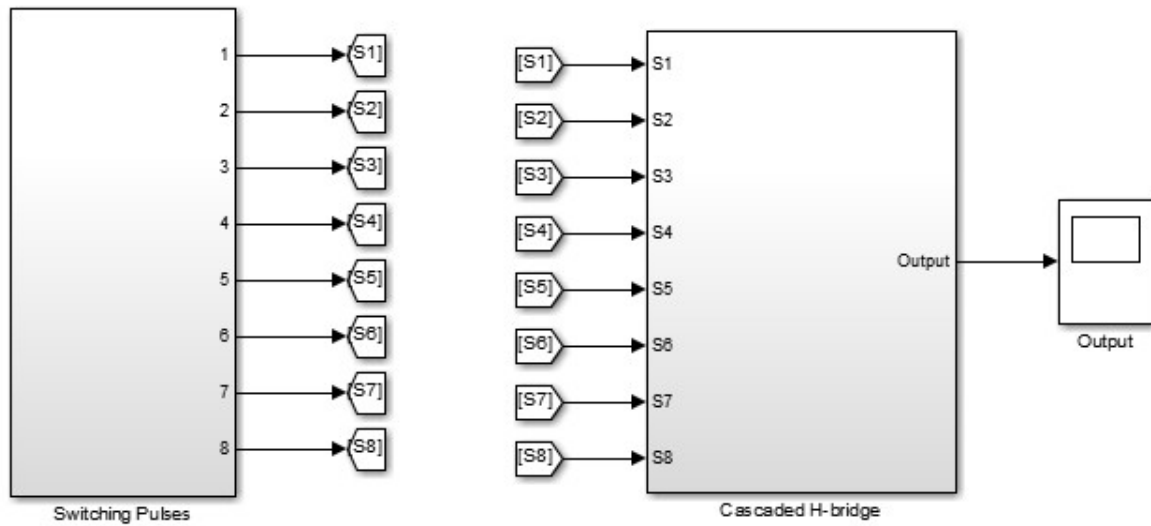


Fig. 3.9: Overall inverter Subsystem